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CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 87 (2024) 102747

Contents lists available at ScienceDirect

Calphad

journal homepage: www.elsevier.com/locate/calphad

Thermodynamic assessment of the Ce-Rh system by the combination of ab-initio calculations and the CALPHAD approach

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ARTICLE INFO

Handling Editor: Prof. Z.K. Liu

Keywords:
Ab-initio calculations
Ce–Rh binary system
VASP code
Associate model equation
Thermodynamic assessment
CALPHAD approach

ABSTRACT

The standard enthalpies of formation in this work have been ~~measured~~ calculated by First -principles calculations within DFT (density functional theory) for the metal compounds in the Ce-Rh system. Thermodynamic data and phase diagram information obtained from the literature were used as the input for a CALPHAD-type optimization and Thermo-Calc software of the Ce-Rh.

The Ce-Rh phase diagram contains seven intermetallic compounds: Ce_7Rh_3, Ce_5Rh_3, Ce_3Rh_2, Ce_5Rh_4, $CeRh$, $CeRh_2$ and $CeRh_3$. All these phases were stoichiometric; The associated model concerning the phase equilibria, could accurately describe the full compositional range and the thermodynamic data input with $CeRh$ as associate. The thermodynamic parameters of all intermetallic compounds were modeled by the Neumann-Kopp principle. The value of the temperature-dependent contributions to the individual Gibbs energies was used for all compounds. A fairly good agreement with all values of the enthalpies of formation calculated in this work using the VASP and data available in the literature. ultimately, a set of self-consistent thermodynamic parameters for the Ce-Rh system was obtained.

1. Introduction

Rare earth alloys with transition elements exhibit interesting properties for high -performance structural applications. They also exhibit a magnificent combination of optical, thermal, electrical, and magnetic properties, which are the subject of a very wide field of construction of new materials, such as the material for hydrogen storage [1,2], permanent magnets [3] and cryogenic magnetic refrigeration [4]. Since Cerium and Rhodium are found to be widely used in the medical field, the alloy of these two elements presents both a very interesting industrial demand, that is to say, to manufacture catalysts for oxidation in humid air [5] or other materials. The low surface free energy favors surface sites for Cerium and Cerium-transition metal catalyst systems are important for serval chemical reactions [6–8]. In particular, Cerium may be used as an ingredient in metal hydride catalysts [9], in the synthesis of ammonia.

Binary transition metal alloy systems have many intermediate phases that are of great interest both to industry and to solid-state theoretical studies. ~~In recent works~~ In this study, the first thermodynamic modelling of the Ce-Rh binary system has been performed and presented. Our results are compared with some published thermochemical data, the predicted values by the Miedema model [10] and values calculated by the ab-initio techniques available in online databases like: Materials project [11], OQMD [12] and Jarvis [13], and databases from Machine Learning models for the formation energies of compounds such as ALIGNN Jarvis [14] and ALIGNN MP [15], SIPFENN [16,17].

No thermodynamic assessment by CALPHAD approach [18] of the Ce-Rh system has been reported in the literature. A thorough assessment

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https://doi.org/10.1016/j.calphad.2024.102747
Received 2 April 2024; Received in revised form 31 August 2024; Accepted 8 September 2024
Available online 26 September 2024
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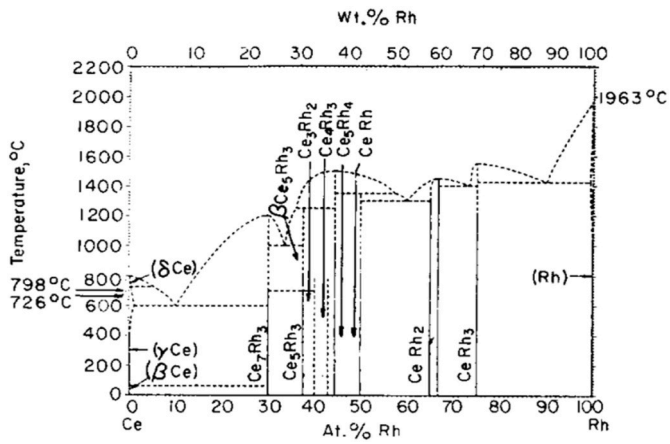

Fig. 1. Experimental phase diagram of the Ce-Rh system estimated by Ref. [19].

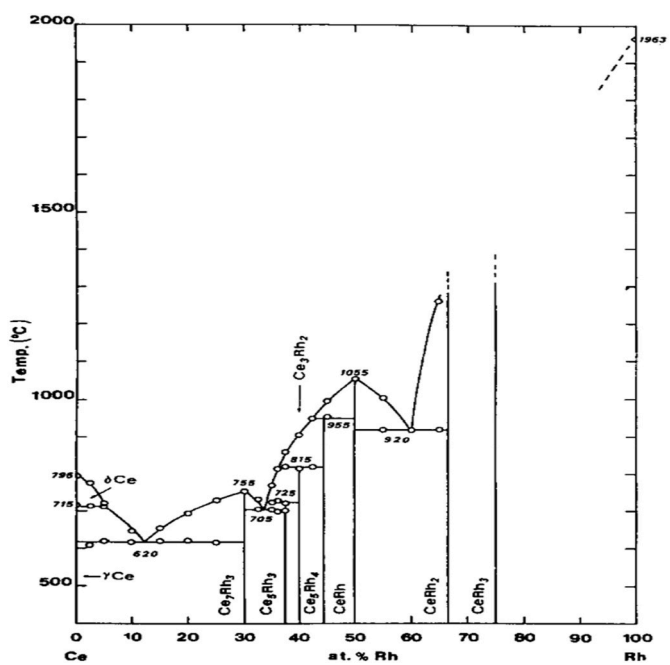

Fig. 2. Experimental phase diagram of the Ce-Rh binary system measured by Ref. [21].

of the Ce-Rh system is necessary to construct reliable thermodynamic descriptions of the Ce-based systems.

2. Literature review

The Ce-Rh phase diagram proposed by Massalski et al. [19] which essentially considers the work of ~~Blazina et al.~~ [20] Singh et al. [20], taking into account its similarity to the Nd-Rh system. According to Refs. [19–25], $CeRh_2$ exhibits slight deviations from stoichiometry, ranging from 0.66 at %Rh to 0.68 at % Rh. This behaviour has been commonly observed for the $LnRh_2$ phase.

Furthermore, pure cerium undergoes a complex series of transitions with increasing temperature: α (fcc, low volume) below 250 °C, β (cP4) above 250 °C and below 590 °C, γ (fcc, high volume) from 590 to 700 °C, and δ (bcc) above the melting point of 700 °C–799 °C. Pressure-induced transformation in pure cerium is described as follows: γ (fcc) - α (fcc) at 0.8 GPa, α'' (monoclinic bcc) at 5 GPa, and α'' (monoclinic bcc) - ε centered tetragonal at 12 GPa. All high-pressure structures can be described as a distorted fcc [26].

The updated version has been investigated by Palenzona et al. [21] shows Fig. 2 who studied the system from 0 to 70 at% Rh. Seven compounds are formed in this system Ce_7Rh_3, Ce_5Rh_3, Ce_3Rh_2 , Ce_5Rh_4, $CeRh$, $CeRh_2$ *and* $CeRh_3$. ~~Their existence was already known and confirmed in this study~~ Their existence has previously been known and confirmed. Previously reported compounds Ce_4Rh_3 and $\beta - Ce_5Rh_3$ ~~and~~ have been excluded. The shape of the diagram is very different from what is reported in the current literature from Refs. [19,21], with melting points in the range of 20–60 at% Rh that are several hundred degrees Celsius lower. The rhodium-rich side of the system could not be studied due to the alloys reacting with the container material at temperatures above 1350 °C.

Three compounds, $CeRh_3$, $CeRh_2$ *and* Ce_5Rh_4, melt incongruently at 998 K, 1088 K, and 1228 K, respectively, while Ce_7Rh_3, $CeRh$, $CeRh_2$ *and* $CeRh_3$. melt congruently at 1028 K, 1328 K, approximately 1700 K, and approximately 1800 K, respectively. At least three eutectic points occur at 12.5 at% Rh and 893 K, 34 at% Rh and 978 K, and 60 at% Rh and 1193 K.

The solubility of rhodium in cerium appears to be very limited as no significant effect on the transformation temperature was observed. The solubility of cerium in rhodium was not studied due to the high melting points of the corresponding alloys [21].

We have essentially based our calculations on the experimental version established by A Palenzona [21],No solubility of Rh in Ce was observed or reported for δCe (see Fig. 2).

Selhaoui and Kleppa [27] measured the enthalpies of the formation of some 5d cerium compounds including Ce-Rh intermetallic phases by direct synthesis calorimetry at hight temperatures (1200 ± 20 °C).The measured standard enthalpies of formation of $CeRh_3$, $CeRh_2$ *and* Ce_5Rh_4 phases are -55.2 ± 1.6 Kj.$mol^{-1}at^{-1}$; -66.5 ± 2.3Kj.$mol^{-1}at^{-1}$ and -70.5 ± 1.9 Kj.$mol^{-1}at^{-1}$ respectively.It is worth nothing that the predicted values from Miedema's Semiempirical equations [10] are less exothermic: -58.5 *Kj* ; -71.3 *Kj*; -69.7 *Kj* for $CeRh_3$, $CeRh_2$ *and* Ce_5Rh_4 Phases,respectively.

In View of the above-mentioned controversies and some inconsistencies revealed by the literature review, the thermodynamic parameters of the Ce-Rh system have been assessed by us through the combination of recent experimental results from Ref. [21] in the range between 0 and 70 at% Rh with those of Massalski et al. [19] in the range from 70 to 100 at %Rh.

The current work is based on the Ce-Rh experimental phase diagram version established by A. Palenzona et al. [21] in the range from 0 to 70 at. %Rh. This was achieved through the utilisation of differential thermal analysis, micrographic analysis, X-ray diffraction and electron microscopy. Similarly, to obtain further details regarding the 70 at. % Rh, up to 100 at. % Rh composition, we have adopted the revised version proposed by Massalski et al. [19], which essentially considers the work of Singh et al. [20], taking into account its similarity to the Nd-Rh system.

Furthermore, the temperature of 1450 °C has been adopted as being the $CeRh_2$ melting thetemperature [22] instead of 1480 °C [19] which is below the temperature of the $liq \leftrightarrow CeRh_2 + CeRh_3$ eutectic reaction (Fig. 1).

Recently, K. T. Jacob et al. [28] conducted an experimental study on the isothermal section at a temperature of 1200 K of the Ce-Rh-O ternary system. The phases mentioned above were confirmed. Furthermore, the crystallographic data of all intermediate compounds formed in the cerium-rhodium system are illustrated in Table 1.

3. First principles calculation

As depicted in Table 1, to address the disparities found in the existing literature data, The crystal structures of all six compounds have been rigorously optimized. This optimization process has paved the way for

Table 1
Crystallographic data for the intermediate phases of the Ce-Rh system.

Compound	Structure Type	Lattice constants (Å)			Notes	Reference
		a	b	c		
Ce_7Rh_3	Th_7Fe_3 ($hP20-P6_3$)	10.051	_	6.378	P^a	[21]
		10.005	_	6.356		[23]
		10.023	_	6.376		[29]
		9.836		6.151		This work
Ce_5Rh_3	Pu_5Rh_3 ($tP32-P4/ncc$)	11.260	_	6.450	P, SC	[21]
Ce_3Rh_2	Er_3Ni_2 ($hR45-R3$)	8.858	_	16.794	P, SC	[21]
		8.835	_	16.76		[30]
		8.739		16.505		This work
Ce_5Rh_4	Sm_5Ge_4 ($oP36-Pnma$)	7.467	14.841	7.624	P	[21]
		7.434	14.86	7.604		[22]
		7.319	14.995	7.499		This work
$CeRh$	CrB ($oC8-Cmcm$)	3.845	10.964	4.174	P	[21]
		3.852	10.986	4.152		[31]
		3.855	10.966	4.153		[32]
		3.857	11.546	3.998		This work
$CeRh_2$	$MgCu_2$ ($cF24-Fd3m$)	7.545	_	_	P^b	[21]
		7.543	_	_	P^c	[21]
		7.538	_	_	Ce-rich	[24]
		7.547	_	_		[24]
		7.534	_	_	Rh-rich	[24]
		7.550	_	_		[33]
		7.539	_	_		[34]
		7.568	_	_		This work
$CeRh_3$	$AuCu_3$ ($cP4-Pm3m$)	4.022	_	_	P^c	[21]
		4.023	_	_		[35]
		4.012	_	_		[23]
		4.020	_	_		[24]
		4.024	_	_		[36]
		4.079	_	_		This work

P, Powder diffraction; SC, single-crystal diffraction. P^a Prepared as Ce_3Rh composition.

Table 2
Thermodynamic parameters calculated in this work.

phases	Thermodynamic models	Parameters
Liquid	$(Ce, CeRh, Rh)_1$	${}^0L^{Liq}_{CeRh} = -176424 + 35.33*T$
		${}^0L^{Liq}_{Ce,CeRh} = +27973 - 114.76*T$
		${}^1L^{Liq}_{Ce,CeRh} = -3793 + 45.46*T$
		${}^0L^{Liq}_{CeRh,Ce} = -78725 - 82.94*T$
		${}^1L^{Liq}_{CeRh,Ce} = -31561 - 11.73*T$
		${}^2L^{Liq}_{CeRh,Ce} = +47737 - 16.26*T$
Fcc_A1	(Ce)	*No excess term optimized*
Fcc_A1	(Ce, Rh)	*No excess term optimized*
Ce_7Rh_3	$(Ce)_{0.7} : (Rh)_{0.3}$	$G^{Ce_7Rh_3}_{Ce,Rh} = 0.7*{}^0G^{Fcc}_{Ce} + 0.3*{}^0G^{Fcc}_{Rh} - 50158 - 6.43*T$
Ce_5Rh_3	$(Ce)_{0.625} : (Rh)_{0.375}$	$G^{Ce_5Rh_3}_{Ce,Rh} = 0.625*{}^0G^{Fcc}_{Ce} + 0.375*{}^0G^{Fcc}_{Rh} - 59907 - 6.20*T$
Ce_3Rh_2	$(Ce)_{0.6} : (Rh)_{0.4}$	$G^{Ce_{16}Ru_9}_{Ce,Rh} = 0.6*{}^0G^{Fcc}_{Ce} + 0.4*{}^0G^{Fcc}_{Rh} - 63173 - 5.95*T$
Ce_5Rh_4	$(Ce)_{0.555} : (Rh)_{0.445}$	$G^{Ce_4Ru_3}_{Ce,Rh} = 0.555*{}^0G^{Fcc}_{Ce} + 0.445*{}^0G^{Fcc}_{Rh} - 69180 - 4.23*T$
$CeRh$	$(Ce)_{0.5} : (Rh)_{0.5}$	$G^{CeRh}_{Ce,Rh} = 0.5*{}^0G^{Fcc}_{Ce} + 0.5*{}^0G^{Fcc}_{Rh} - 73672 - 3.22*T$
$CeRh_2$	$(Ce)_{0.33} : (Rh)_{0.67}$	$G^{CeRh_2}_{Ce,Ru} = 0.33*{}^0G^{Fcc}_{Ce} + 0.67*{}^0G^{Fcc}_{Rh} - 70037 - 5.70*T$
$CeRh_3$	$(Ce)_{0.25} : (Rh)_{0.75}$	$G^{CeRh_3}_{Ce,Ru} = 0.25*{}^0G^{Fcc}_{Ce} + 0.75*{}^0G^{Fcc}_{Rh} - 63041 - 2.82*T$

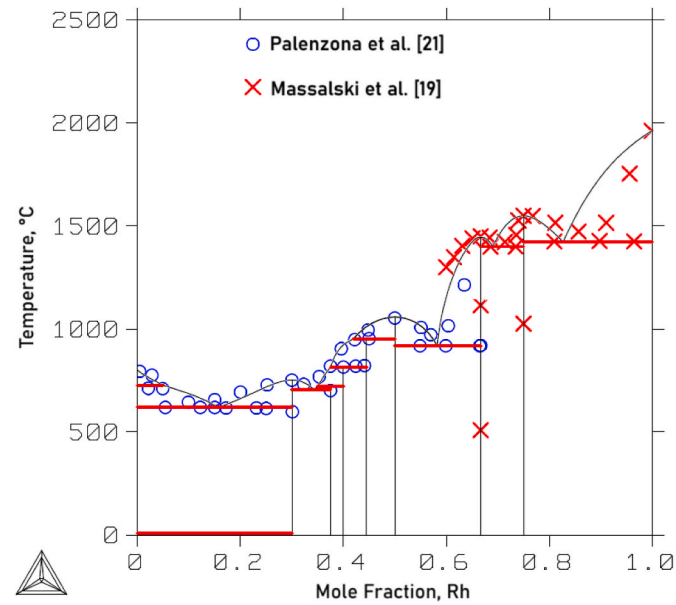

Fig. 3. The phase diagram of the Ce-Rh binary system optimized in the present work compared with experimental data of the literature [19,21].

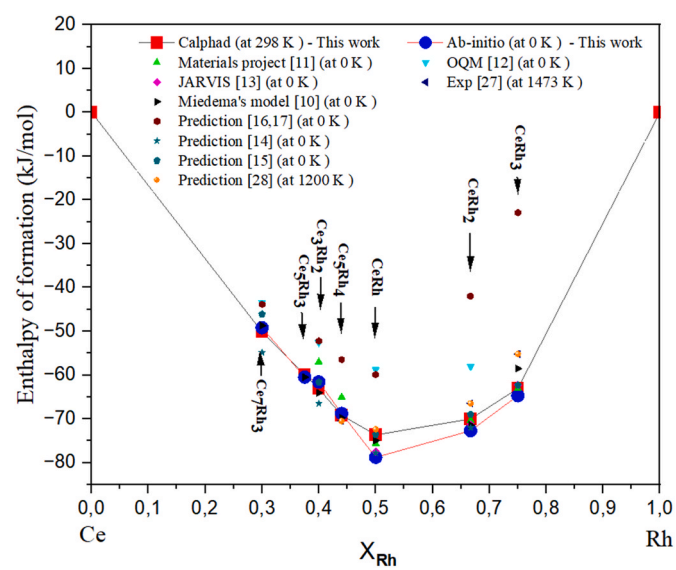

Fig. 4. The enthalpies of formation calculated in this work using the Calphad approach compared with the results measured in the present work by calorimetry [27] and with the values predicted by Refs. [10–17,28].

us to initiate first-principles calculations of the enthalpies of formation.

The formation enthalpies for Ce-Rh structures from the results of ab initio calculations were estimated as follows:

$$\Delta_f H = E^{Ce_xRh_y}_{total} - \frac{x}{x+y} \times E^{Ce-Fcc_A1}_{total} - \frac{y}{x+y} \times E^{Rh-Fcc_A1}_{total} \quad (1)$$

Where $E^{Ce_xRh_y}_{total}, E^{Ce-Fcc_A1}_{total}, E^{Rh-Fcc_A1}_{total}$ are the ab-initio total energies calculated within the density functional theory (DFT) [37] at 0 K using generalized-gradient approximation (GGA) [38] for the exchange-correlation energy in the Perdew-Burke-Ernzernhof (PBE) functional form and projector augmented wave (PAW) method [39] for accounting core-valence electron interactions.

Ab initio calculations were performed in the VASP package [40,41]. We used Ce and Rh_pv PAW potentials that were supplied within the

Table 3
Comparison between the calculated invariant reactions and the experimental data from the literature.

Invariant reactions	Reaction type	*This work*		literature [21]		literature [19]		Prediction data [47,48]
		$x(Rh)/at$	$T(K)$	$x(Rh)/at$	$T(K)$	$x(Rh)/at$	$T(K)$	$T(K)$
$Liq \Leftrightarrow (\delta Ce)$	melting	0	1066	0	1066	–	–	1092
$Liq + (\delta Ce) \Leftrightarrow (\gamma Ce)$	Metatectic	0.05	988	0.051	988	–	–	–
$Liq + (\delta Ce) \Leftrightarrow Ce_7Rh_3$	*Eutectic*	0.1567	892.8	0.125	893	–	–	–
$Liq \Leftrightarrow Ce_7Rh_3$	*Congruent*	0.3	1025	0.3	1028	–	–	1088
$Liq + Ce_7Rh_3 \Leftrightarrow Ce_5Rh_3$	*Eutectic*	0.3430	979.8	0.34	978	–	–	–
$Liq + Ce_5Rh_3 \Leftrightarrow Ce_3Rh_2$	*peritectic*	0.3498	993.8	0.35	998	–	–	–
$Liq + Ce_3Rh_2 \Leftrightarrow Ce_5Rh_4$	*peritectic*	0.3750	1087	0.366	1088	–	–	–
$Liq \Leftrightarrow Ce_5Rh_4 + CeRh$	*peritectic*	0.4196	1226	0.426	1228	–	–	–
$Liq \Leftrightarrow CeRh$	Congruent	0.5	1331	0.5	1328	–	–	1352
$Liq \Leftrightarrow CeRh + CeRh_2$	*Eutectic*	0.5821	1191	0.600	1193	–	–	–
$Liq \Leftrightarrow CeRh_2$	Congruent	0.667	1720	0.667	–	0.667	1723	1668
$Liq \Leftrightarrow CeRh_2 + CeRh_3$	*Eutectic*	0.6907	1674	–	–	0.728	1673	–
$Liq \Leftrightarrow CeRh_3$	Congruent	0.75	1822	–	–	0.75	1823	1773
$Liq \Leftrightarrow (\alpha Rh) + CeRh_3$	Eutectic	0.8271	1696	–	–	0.898	1698	–
$Liq \Leftrightarrow (\alpha Rh)$	Melting	1	2236			1	2236	1657

VASP package [40,41]. The energy cut-of (520 eV) and k-point density (the maximal distance between the two nearest k-points in the reciprocal space was less than 0.1 2*PI/Angstrom) were chosen to ensure the 0.001 eV/atom total energy convergence.

We have re-evaluated the total energy calculations for intermetallic compounds using a convergence threshold for forces of 0.001 eV/Å. The recalculated energies show a maximal deviation of only 2 % from those obtained with the previous criterion of 0.05 eV/Å, indicating no major difference that would affect our conclusions.

All structures were relaxed until the forces acting on atoms became less than 0.05 eV/A.

The total energies of the compounds and their constituent elements in their most stable states were determined by interpolating energy-volume curves using cubic splines. These energy-volume curves were generated by relaxing cell shapes and atomic positions at various fixed volumes.

In this work, cubic splines fitting was used to get the total energy of the equilibrium state of each pure element structure and intermetallic compound, with the aim of optimising all structures.

Since the influence of the pressure term on the condensed phases was not accounted for, the total energy was calculated at 0 K, the calculated energy of formation in this study is considered as the enthalpy of formation at 298 K. Furthermore, throughout the optimization process,we have been introduced the measured temperature to each determined accessible enthalpy of formation. These calculated values should be juxtaposed with experimental results for comparison.

4. Thermodynamic modelling

The Ce-Rh system includes the following phases: liquid, solid solution phases (Rh), (αCe), (βCe), and (δCe), and seven stoichiometric compounds, Ce_7Rh_3, Ce_5Rh_3, Ce_3Rh_2, Ce_5Rh_4, $CeRh$, $CeRh_2$ and $CeRh_3$.

4.1. Pure elements

The solution phases concerning liquid, solid solution phases, (Rh), (αCe), (βCe), and (δCe) were described by the substitutional solution model, The Gibbs energy function $G_i^{\varphi}(T) = {}^0G_i^{\varphi} - H_i^{SER}$ (298.15 K) for the element i ($i = Ce, Rh$) in the phase ($Fcc_A1, Bcc_A2,$ and liquid phase) is given by the following formula:

$$G_i^{\varphi}(T) = a + bT + cT + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9} \quad (2)$$

where a, b, c, d, e, f, g, and h are the coefficients to be optimized. Moreover H_i^{SER} (298.15 K) is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, fcc_A1 for Ce and Rh and T is the absolute temperature.

In this work, the Gibbs energy functions of the pure elements Ce and Rh are taken from the compilation of the SGTE (Scientific Group Thermo data Europe) of Dinsdale [42].

The Gibbs energy of the element i, in its SER state, is denoted by $GHSER_i$ as follows:

$$GHSER_{CE} = G_{Ce}^{FCC_A1} - H_{Ce}^{FCC_A1}(298.15K) \quad (3)$$

$$GHSER_{Rh} = G_{Rh}^{FCC_A1} - H_{Rh}^{FCC_A1}(298.15K) \quad (4)$$

4.2. Associate model

The choice of CeRh compound as an associate can be assessed from two aspects. One observation is that a steep liquidus around the CeRh intermetallic phase indicates high thermodynamic stability in this composition. The other observation is that the mixing enthalpies measured for the liquid phase show a significant change at around 50 at. % Rh. Hence, the Gibbs ~~free~~ energies of the binary liquid for a 1-mol species are given:

$$G_m^{\varphi} = {}^{ref}G_m^{\varphi} + {}^{id}G_m^{\varphi} + {}^{xs}G_m^{\varphi} + {}^{f}G_m^{\varphi} \quad (5)$$

The association model is also based on the theory that a complex is being formed in the liquid phase. $pA + qB \rightleftharpoons ApBq$ (Where $ApBq$ corresponds to the C complex). A *and* B are the pure components initially in the mixture; p and q are the stoichiometric coefficients. Therefore, the various components of the Gibbs energy can be represented by the following equations:

$$^{ref}G_m^{\varphi} = \left\{x_A\left[{}^0G_A^{liq} - H_A^{SER}(298.15)\right] + x_B\left[{}^0G_B^{liq} - H_B^{SER}(298.15)\right]\right\}(y_A + (p+q)y_B + y_C) \quad (6)$$

$$^{id}G_m^{\varphi} = RT\left(\sum_{i=A}^{C} y_i \ln y_i\right) \quad (7)$$

$$^{f}G_m^{\varphi} = \left(y_A^0 G_A^{Liq}\right) + \left(y_B^0 G_B^{Liq}\right) + \left(y_C^0 G_C^{Liq}\right) \quad (8)$$

where R is the constant of the ideal gas; T is the absolute temperature; A, B, C are the three species in the liquid; y_A, y_B *and* y_C their site fractions normalized to $y_A + y_B + y_C = 1$ and x_A, x_B the initial molar fraction of the solution, without considering the associated species, normalized to $x_A + x_B = 1$. The excess Gibbs energy can be represented by the Redlich-

Table 4
Comparison between the optimized thermodynamic parameters of the intermetallic compounds derived in this work and those drawn from the literature.

Phases	$\Delta_f H$ (KJ/mol. At)	Temperature (K)	$\Delta S\left(J.K^{-1}mol^{-1}at^{-1}\right)$	Technique used	Reference
Ce_7Rh_3	− 50.158	298	− 6.43	Optimization	This work
	− 48.681	0	–	Prediction	[10]
	− 43.416	0	–	Ab-initio	[12]
	− 54.85828	0		Prediction	[14]
	− 46.078848	0		Prediction	[15]
	− 43.85981	0		Prediction	[16,17]
	− 49.200	0	–	Ab-initio	This work
Ce_5Rh_3	− 59.907	298	− 6.20	Optimization	This work
	− 60.474	0	–	Prediction	[10]
Ce_3Rh_2	− 63.173	298	− 5.95	Optimization	This Work
	− 64.066	0	–	Prediction	[10]
	− 57.019	0	–	Ab-initio	[11]
	− 52.5816	0	–	Ab-initio	[12]
	− 66.532608	0		Prediction	[14]
	− 61.631424	0		Prediction	[15]
	− 52.22462	0		Prediction	[16,17]
	− 61.600	0	–	Ab-initio	This work
Ce_5Rh_4	− 69.180	298	− 4.23	Optimization	This work
	− 70.50	1200	− 10.22	Prediction	[28]
	− 70.50	1473 ± 2	–	Calorimetry	[27]
	− 69.790	0	–	Prediction	[10]
	− 65,027	0	–	Ab-initio	[11]
	− 56.47599	0		Prediction	[16,17]
	− 68.800	0	–	Ab-initio	This work
$CeRh$	− 73.672	298	− 3.22	Optimization	This work
	− 72.40	1200	− 10.5	Prediction	[28]
	− 74.956	0	–	Prediction	[10]
	− 75,640	0	–	Ab-initio	[11]
	− 58,756	0	–	Ab-initio	[12]
	− 77,473	0	–	Ab-initio	[13]
	− 77.840064	0		Prediction	[14]
	− 73.594944	0		Prediction	[15]
	− 59.90443	0		Prediction	[16,17]
	− 78.800	0	–	Ab-initio	This work
$CeRh_2$	− 70.037	298	− 5.70	Optimization	This work
	− 66.50	1200	− 9.64	Prediction	[28]
	− 66.50	1473 ± 2	–	Calorimetry	[27]
	− 71.386	0	–	Prediction	[10]
	− 70,333	0	–	Ab-initio	[11]
	− 58,080	0	–	Ab-initio	[12]
	− 71,298	0	–	Ab-initio	[13]
	− 72.01267	0		Prediction	[14]
	− 68.944608	0		Prediction	[15]
	− 41.97845	0		Prediction	[16,17]
	− 72.700	0	–	Ab-initio	This work
$CeRh_3$	− 63.041	298	− 2.82	Optimization	This work
	− 55.20	1200	− 8.01	Prediction	[28]
	− 55.20	1473 ± 2	–	Calorimetry	[27]
	− 58.517	0	–	Prediction	[10]
	− 63,097	0	–	Ab-initio	[11]
	− 62,229	0	–	Ab-initio	[13]
	− 62.191008	0	–	Prediction	[14]
	− 62.306784	0	–	Prediction	[15]
	− 22.90435	0	–	Prediction	[16,17]
	− 64.700	0	–	Ab-initio	This work

Kister polynomial.

$$^{XS}G_m^{\varphi} = y_i y_j \left(\sum_{\lambda=0}^{\lambda} L_{i,j}^{Liq} \left(y_i - y_j \right)^{\lambda} \right) \tag{9}$$

Where *i and j* correspond to species A, B or C. The interaction parameter $^{\lambda}L_{i,j}^{Liq}$ is a function of the temperature.

For the liquid phase, the associated solution model was chosen. This model assumes the existence of molecular-like groups of atoms as the manifestation of intense interatomic interactions. These groups (associates) are expected to define the structure and the thermodynamics of the system and should be especially important at low temperatures when the role of interatomic interactions increases. Besides, it is helpful to describe systems with pronounced irregularities on the curves of dependence of thermodynamic functions on composition.

4.3. Stoichiometric intermetallic compounds

In this system there are seven stoichiometric compounds of the A_pB_q type : Ce_7Rh_3, Ce_5Rh_3, Ce_3Rh_2, Ce_5Rh_4, $CeRh$, $CeRh_2$, and $CeRh_3$. The Gibbs ~~free~~ energy of these is expressed by the following expression:

$$^{0}G_{Ce_pRh_q} = \frac{p}{p+q}\,^{0}G_{Ce} + \frac{q}{p+q}\,^{0}G_{Rh} + a + bT + \mathrm{cTln(T)} \tag{10}$$

With $^{0}G_{Ce}$ and $^{0}G_{Rh}$ are the Gibbs energy of the pure elements Ce and Rh, respectively.

a and b are two parameters to be adjusted in this work based on available information in the literature and our ab-initio calculation.

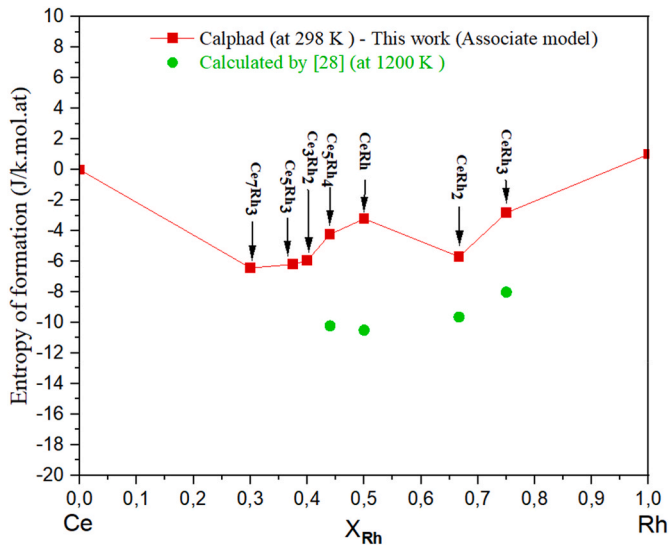

Fig. 5. Entropies of formation of intermetallic compounds calculated in this work compared with literature data of Jacob et al. [28].

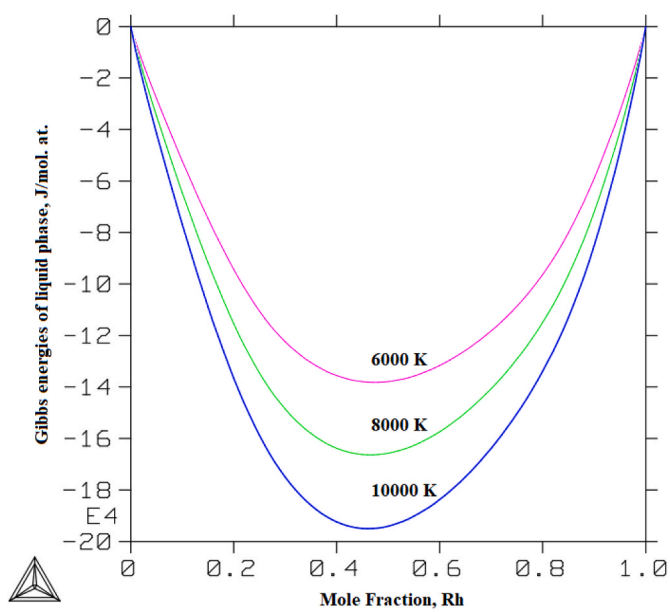

Fig. 6. The enthalpy of mixing of the liquid phase calculated at different temperatures (6000-8000-10000 K) using the association solution model [42.43].

In most cases, the following formulas are employed for the purpose of calculations,

$$\Delta H(T=1473K)=\Delta H(T=0K)+\int_0^{1473} C_p dT$$

$$\Delta H(T=1473K)=\Delta H(T=0K)+\int_0^{1473} [a+bT+cTlnT]dT$$

$$\Delta H(T=1473K)=\Delta H(T=0K)+a[T]_0^{1473}+b\left[\frac{T^2}{2}\right]_0^{1473}$$
$$+c\left[\frac{T^2}{2}ln(T)-\frac{1}{4}T^2\right]_0^{1473}$$

However, The T*lnT term comes from the treatment of the

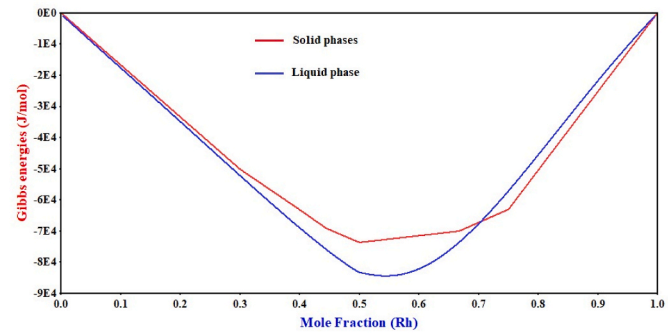

Fig. 7. Gibbs energies of the solid phases and liquid phase in the Ce-Rh system at T = 298.15 K using associate model.

experimental Cp data for a phase. Due to the scarcity of data on heat capacities. Thus, a theoretical calculation based on the Neumann-Kopp rule was applied.

5. Procedure of the thermodynamic modelling

In this section, the approach to the thermodynamic modelling of the studied systems is described. The optimization of the thermodynamic parameters of Ce-Rh system was carried out using PARROT module of Thermo-Calc software, developed by Jansson [43] and Sundman et al. [44].

The first step involved initiating the model with the liquid phase, utilizing zero-order interaction parameters as initially undertaken. Subsequently, intermetallic compounds were systematically introduced into the model. During optimization, first and second-order interaction parameters of the liquid phase were incorporated to accurately describe the liquidus shape in the phase diagram.

The liquid phase was modeled as a substitutional solution using an associate Model [45,46] in our optimization process to prevent the occurrence of an undesired inverted miscibility gap in the solution phases at elevated temperatures during the phase diagram calculation. In this system, the calculated invariant temperatures were adjusted iteratively to minimize the sum of the squares of errors between the experimental data and computed values until most of the selected experimental information was reproduced within expected uncertainty limits. The weights were systematically adjusted to enhance the overall consistency of the data. The resulting thermodynamic parameters are listed in Table 2.

6. Optimization and calculated results

Prior to initiating the thermodynamic modelling of the Ce-Rh system using the CALPHAD approach, a comprehensive examination was conducted on all confirmed stable intermetallic compounds within this system, each possessing known structural characteristics. The resulting tables (Tables 1 and 2) and Fig. 4 present the lattice parameters and enthalpies of formation, as calculated through the Ab-initio method within this study.

In this work, cubic splines fitting was used to get the total energy of the equilibrium state of the intermetallic compounds.

Redlich-Kister polynomials have commonly been used to represent energy functions. The global nature of these polynomials often prevents them from capturing local phenomena. Spline functions present an effective alternative for representing chemical potential and energy data. Splines are piecewise polynomials with a specified order of continuity across the entire domain. Because of their piecewise structure, they are able to accurately represent local features of the data. The inherent constraints prescribing that the spline and a certain number of its derivatives be continuous across all subdomain junctions give splines an order of global continuity. These local and global properties make

splines an important tool in curve fitting. However, our thermodynamic description for the present system does not take the change in the pressure.

The calculated phase diagram of the Ce-Rh binary system using CALPHAD method [14] and PARROT module of the Thermo-Calc software [43,44]. This software is used to introduce a consistent set of experimental thermodynamic data into the optimization.

Fig. 3 displays our calculated phase diagram compared to experimental data [19,21] and compiled data taken from Figs. 1 and 2. It is noted that the calculated phase diagram is in good agreement in the Ce-rich part and reasonable agreement was noted over the compositional range of 70–100 at. %Rh with the experimental diagrams from the literature. However, a small difference is noted for the calculated liquidus for the eutectic reaction (($Liq \Leftrightarrow Rh + CeRh_3$) which is slightly different from the diagram proposed by Massalski [19]. Due to the lack of experimental studies, and from the point of view of the symmetry of the liquidus around the congeneric melts, the results calculated in this work are considered acceptable. Table 3 presents all the invariant reactions calculated in the Ce-Rh system, compared with literature data [19,21]. These values are also compared with those predicted by a graph neural network model [47,48], showing reasonable agreement and a best consistency with experimental data of Palenzona et al. [21], was achieved over the compositional range from 0 to 70 at. %Rh. A discrepancy is less than 3K for invariant temperatures and 3 at. % for atomic compositions. However, it should be noted that some predicted results have a relatively short standard error, making this comparison acceptable.

The enthalpies of formation for seven intermetallic compounds existing in the Ce-Rh system are presented in Table 4 and Fig. 4. These values are compared with experimental data obtained from calorimetry by Selhaoui et al. [27], data calculated by Miedema et al. [10], tree ML predictions (by as ALIGNN Jarvis [14],ALIGNN MP [15] and SIPFENN [16,17]),our optimized values, and those calculated using ab-initio methods in this study. Remarkably, a high level of agreement is observed between our calculated data and the experimentally measured results, as well as with the theoretically predicted values by the model of Miedema et al. [10].

It exists the inconsistencies concern the experimental enthalpies of formation measured by Selhaoui and Kleppa [27] at 1473 K using DSC and ones predicted by Miedema model. This later, gives more exothermic values. From Table 4 and Fig. 4, it is evident that a satisfactory consistency between our calculated by ab-initio and experimental data from Ref. [27], as well as the theoretical predicted data by the Miedema et al. [10].

The machine learning results are in good agreement with those obtained from DFT. ALIGNN Jarvis [14] predicted 5 out of 6 configurations, missing only one, while SIPFENN [16,17] predicted all 6 configurations. However,upon examining the results, SIPFENN [16,17] shows a significant difference at 75.0 at. % Rh in $CeRh_3$ compared to DFT results. For Alignn [14,15], the maximum difference is 5.658 kJ/mol.atom at 30.0 at. % Rh in Ce_7Rh_3.

Overall, the observed differences are minor and can be attributed to the various electronic structure methods and approximations used by different research groups. Consequently, the enthalpy of formation values obtained from DFT were used as excess terms in the Redlich-Kister polynomial, significantly simplifying the thermodynamic modeling and enhancing the theoretical robustness of the thermodynamic description.

The results of the machine learning demonstrate significant potential in identifying stable configurations and providing accurate predictions of ΔH_{form} for each phase. This reduces the computational effort required by DFT-based calculations.

Fig. 5 illustrates the entropies of formation for the solid compounds calculated in this study using the associates' model [45,46], compared with values calculated by K.T. Jacob et al. [28]. It is noteworthy that our calculated values exhibit higher positivity compared to those predicted by Jacob et al. [28]. Due to the absence of experimental data, further comparisons are not feasible.

In order to determine the stability of the liquid phase as a function of temperature, the liquid stability at high temperatures has been plotted. Fig. 6 shows the Gibbs energy curve for the mixing of the liquid phase at 6000 K, 8000 K and 10000 K calculated in this work by the associate model [45,46]. In this figure, it is observed that the liquid is very stable even at high temperatures. ~~Therefore, no reverse miscibility gap formation was detected at high temperatures in our calculation~~. Just to know the objective of this study is to validate the Gibbs energy model from a mathematical perspective.

In Fig. 7, the phenomenon of liquid re-stabilization at low temperatures using the Associate model was investigated by comparing the Gibbs energy of solid phases (represented by the red line) to that of the liquid phase (represented by the blue line) at 298.15 K. According to Schmid-Fetzer et al. [49], the occurrence of low-temperature artifacts is more likely at lower temperatures (T-s), but unary data are only available above T = 298.15 K [42]. Therefore, the Gibbs energies of solid phases are compared to that of the liquid phase at this temperature.

It is observed that the Gibbs energy curve for the liquid phase lies within a less positive range compared to the Gibbs energy of the solid phases (which are highly stable) in the composition range of 0–76 at % Rh, and within a more positive range in the range of 76–100 at %Rh. Given the close proximity of the Gibbs energies of the liquid and solid phases at T = 298.15 K in the Ce and Rh-rich regions, it can be concluded that the low-temperature artifact does not occur in this system.

7. Conclusion

The thermodynamic properties of the optimization and phase diagrams within the Ce-Rh system were achieved through a synergistic approach, integrating first-principles calculations and the CALPHAD method. A systematic compilation of reliable literature data, along with outcomes from Ab-initio calculations, was employed to derive a coherent set of thermodynamic parameters. Our assessed phase equilibria closely match experimental observations.

Remarkably, this study marks the initial determination of the enthalpies of formation and entropies for key intermetallic compounds in the Ce-Rh system, namely Ce_7Rh_3, Ce_5Rh_3, Ce_3Rh_2 , Ce_5Rh_4, $CeRh$, $CeRh_2$ *and* $CeRh_3$, as well as for the liquid phase. This data was pivotal in constructing the phase diagram of this system, through amalgamating two diagram versions. In light of these findings, a comprehensive experimental investigation is strongly recommended to validate the results, especially within the Rh-rich composition range.

CRediT authorship contribution statement

H. Bouchta: Writing – original draft, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation. **N. Selhaoui:** Writing – review & editing, Validation, Supervision. **D.O. Poletaev:** Software. **M.A. Boukideur:** Writing – review & editing, Methodology, Funding acquisition. **A. Bendarma:** Validation, Supervision. **S. Kardellass:** Validation. **A. Marjaoui:** Supervision, Software, Data curation. **M. Zanouni:** Software, Resources, Data curation. **A. Khadija:** Methodology, Supervision, Validation

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.calphad.2024.102747.

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Update

Calphad
Volume 87, Issue , December 2024, Page

DOI: https://doi.org/10.1016/j.calphad.2024.102756

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 87 (2024) 102756

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Corrigendum to "Thermodynamic assessment of the Ce-Rh system by the combination of ab-initio calculations and the CALPHAD approach" [Calphad 87 (2024) 102747]

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The authors regret that the following typo is in the original text and should be corrected in this corrigendum:

Page 1: Abstract, paragraph 1.

The standard enthalpies of formation in this work have been calculated by First -principles calculations within DFT (density functional theory) for the metal compounds in the Ce-Rh system. Thermodynamic data and phase diagram information obtained from the literature were used as the input for a CALPHAD-type optimization and Thermo-Calc software of the Ce-Rh.

The Ce-Rh phase diagram contains seven intermetallic compounds: Ce7Rh3 ,Ce5Rh3 ,Ce3Rh2 ,Ce5Rh4, CeRh, CeRh2 and CeRh3. All these phases were stoichiometric; The associated model concerning the phase equilibria, could accurately describe the full compositional range and the thermodynamic data input with as associate. The thermodynamic parameters of all intermetallic compounds were modelled by the Neumann-Kopp principle. The value of the temperature-dependent contributions to the individual Gibbs energies was used for all compounds. A fairly good agreement with all values of the enthalpies of formation calculated in this work using the VASP and data available in the literature. ultimately, a set of self-consistent thermodynamic parameters for the Ce-Rh system was obtained.

Page 1: 1.Introduction, paragraph 2.

Binary transition metal alloy systems have many intermediate phases that are of great interest both to industry and to solid-state theoretical studies. In this study, the first thermodynamic modelling of the Ce-Rh binary system has been performed and presented. Our results are compared with some published thermochemical data, the predicted values by the Miedema model [10] and values calculated by the ab-initio techniques available in online databases like: Materials project [11], OQMD [12] and Jarvis [13], and databases from Machine Learning models for the formation energies of compounds such as ALIGNN Jarvis [14] and ALIGNN MP [15], SIPFENN [16,17].

Page 2: 2. Literature review, paragraph 1.

The Ce-Rh phase diagram proposed by Massalski et al. [19] which essentially considers the work of Singh et al.[20], taking into account its similarity to the Nd-Rh system. According to [19–25], $CeRh_2$ exhibits slight deviations from stoichiometry, ranging from 0.66 at %Rh to 0.68 at % Rh. This behaviour has been commonly observed for the $LnRh_2$ phase.

Page 2: 2. Literature review, paragraph 3.

The updated version has been investigated by Palenzona et al. [21] shows Fig. 2 who studied the system from 0 to 70 at% Rh. Seven compounds are formed in this system Ce_7Rh_3, Ce_5Rh_3, Ce_3Rh_2 , Ce_5Rh_4, $CeRh$, $CeRh_2$ *and* $CeRh_3$. Their existence has previously been known and confirmed. Previously reported compounds Ce_4Rh_3 and $\beta - Ce_5Rh_3$ have been excluded. The shape of the diagram is very different from what is reported in the current literature from [19,21], with melting points in the range of 20–60 at% Rh that are several hundred degrees Celsius lower. The rhodium-rich side of the system could not be studied due to the alloys reacting with the container material at temperatures above 1350 °C.

Page 4: 4.2. Associate model, paragraph 1.

The choice of CeRh compound as an associate can be assessed from two aspects. One observation is that a steep liquidus around the CeRh intermetallic phase indicates high thermodynamic stability in this composition. The other observation is that the mixing enthalpies

DOI of original article: https://doi.org/10.1016/j.calphad.2024.102747.

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https://doi.org/10.1016/j.calphad.2024.102756

Available online 18 October 2024
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measured for the liquid phase show a significant change at around 50 at. % Rh. Hence, the Gibbs energies of the binary liquid for a 1-mol species are given:

Page 5 : 4.3. Stoichiometric intermetallic compounds, paragraph 1.

In this system there are seven stoichiometric compounds of the A_pB_q type : Ce_7Rh_3, Ce_5Rh_3, Ce_3Rh_2, Ce_5Rh_4, $CeRh$, $CeRh_2$, and $CeRh_3$. The Gibbs energy of these is expressed by the following expression:

Page 7: 6. Optimization and calculated results, paragraph 12.

In order to determine the stability of the liquid phase as a function of temperature, the liquid stability at high temperatures has been plotted. Fig. 6 shows the Gibbs energy curve for the mixing of the liquid phase at 6000 K, 8000 K and 10000 K calculated in this work by the associate model [45, 46]. In this figure, it is observed that the liquid is very stable even at high temperatures. Just to know the objective of this study is to validate the Gibbs energy model from a mathematical perspective.

The authors would like to apologise for any inconvenience caused.